Somjit Roy

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Research Interests and Skill Set

My research advances next-generation statistical modeling frameworks that bridge data-driven learning with scientific reasoning. I develop novel methodologies for complex real-world problems in materials discovery, geophysics, bioinformatics, and finance. Working at the intersection of scientific machine learning and Bayesian modeling & computation, I focus on scientifically guided probabilistic inference, physics-informed modeling, approximate Bayes, tree-based models, and Bayesian optimization, with the goal of building interpretable, scalable, and scientifically grounded learning systems.

Core methods: Probabilistic SciML | Symbolic Regression | Approximate Bayes | Causal Inference | Bayesian Optimization

Education and Training

Texas A&M University, College Station, TX, USA — 2023–2028 (expected)
Doctor of Philosophy (Ph.D.) in Statistics — GPA: 4.00/4.00
Advisors: **Bani K. Mallick** (TAMU Statistics) & **Debdeep Pati** (UW–Madison Statistics)
Area of Study: Probabilistic Scientific Machine Learning and Computational Bayes

University of Calcutta, Kolkata, West Bengal, India — 2021–2023
Master of Science (M.Sc.) in Statistics, Ranked First Class First — GPA: 8.15/10.00

St. Xavier's College (Autonomous), Kolkata, West Bengal, India — 2018–2021
Bachelor of Science (B.Sc.) in Statistics, Honours — GPA: 8.63/10.00

Experience

Upstart Network, Inc. — June 2026–Aug 2026
Research Scientist Intern, Machine Learning — Remote | Austin, TX, USA

- Built a **selection-bias-aware NPV-RMSE** framework for funded borrowers, correcting applicant-level performance estimates by 9–11% and enabling more reliable credit-risk model evaluation.
- Designed a **two-stage transport-weighting** scheme to adjust for funding selection bias, improving applicant covariate balance by 36%.
- Trained propensity-score models in `Python` on `AWS` across 5.4M+ loans and 5K+ features to power the reweighting pipeline.

Los Alamos National Laboratory — May 2025–Aug 2025
Graduate Summer Intern — Los Alamos, NM, USA

- Developed a scalable **physics-informed neural operator** for acoustic-wave modeling in complex, heterogeneous media.
- Scaled distributed training with `PyTorch DDP` and `CUDA` across `GPUs` on LANL high-performance compute clusters.
- Generalized wavefield predictions accurately to unseen spatial-temporal coordinates and held-out heterogeneous media.

Tata Electronics Pvt. Ltd. — May 2021–Aug 2021
Data Science Intern — Bengaluru, KA, India

- Analyzed **sandblasting and anodizing trials** to identify process drivers of surface gloss, texture and recommend in-spec parameter settings.
- Applied random forests, support vector machines, and ridge/lasso regression to mitigate multicollinearity and rank parameters by influence.
- Identified band and jig positions as dominant sources of finish variation, cutting production time by two person-days.

Papers and Manuscripts

Tags: ★ Probabilistic SciML ★ Symbolic Regression ☆ Variational Inference ☆ Bayesian Optimization ☆ Quantum Cryptography

6. *Roy, S.*, Dey, P., Pati, D., & Mallick, B. K. (2026). A Generalized Tangent Approximation based Variational Inference Framework for Strongly Super-Gaussian Likelihoods. (*Forthcoming in Journal of the American Statistical Association (JASA), Theory & Methods, accepted*). arXiv ↗ ☆

5. *Roy, S.*, Dey, P., & Mallick, B. K. (2026). VaSST: Variational Inference for Symbolic Regression using Soft Symbolic Trees. *Forty-Second Annual Conference on Uncertainty in Artificial Intelligence*. PMLR ↗ ★ ★ ☆

4. *Roy, S.*, Jaiswal, P., Bhattacharya, A., Pati, D., & Mallick, B. K. (2026+). On Frequentist Regret Analysis of Fractional Gaussian Process Thompson Sampling. (*Submitted*). arXiv ↗ ☆

3. *Roy, S.*, Dey, P., Mallick, B. K., & Pati, D. (2026+). Probabilistic Symbolic Regression for Equation Discovery via Operator-induced and Regularized Symbolic Forests. (*Submitted*). arXiv ↗ ★ ☆
   2026 IMS Hannan Graduate Student Travel Award | 2026 ASA SBSS Student Paper Award | BOSS 2026, 2nd place (TidBit) | TAMIDS Student Travel Grant featured by the Texas A&M College of Arts and Sciences (news ↗)

2. Kumar, A., Maitra, S., & *Roy, S.* (2024). Almost Perfect Mutually Unbiased Bases that are Sparse. *Journal of Statistical Theory and Practice*, **18**, 61. DOI ↗ ☆

1. Chaudhury, S., Kumar, A., Maitra, S., *Roy, S.*, & Sen Gupta, S. (2022). A Heuristic Framework to Search for Approximate Mutually Unbiased Bases. In *Cyber Security, Cryptology, and Machine Learning* (CSCML 2022), Lecture Notes in Computer Science, vol. **13301**. Springer, Cham. DOI ↗ ☆

Open-Source Software

GitHub `VaSST` (*Roy, S.* & Dey, P., 2026); `HierBOSSS` (*Roy, S.* & Dey, P., 2026); `TAVIE-SSG` (*Roy, S.* & Dey, P., 2026)

CRAN `bayesestdft` (*Roy, S.* & Lee, S. Y., 2025); `GoodFitSBM` (Ghosh, S., *Roy, S.*, & Pati, D., 2024)

Technical Skills

Programming & Data	`Python`, `SQL`, `C++`, `R`, `NumPy`, `pandas`
ML & Scientific Computing	`PyTorch`, `PyTorch Lightning`, `JAX`, `TensorFlow`, `scikit-learn`, `SciPy`
Probabilistic Computing	`PyMC`, `NumPyro`, `Stan/RStan`, `mcmc`, `coda`
GPU & Distributed Computing	`CUDA`, `GPU computing`, `PyTorch DDP`, `MPI`, `OpenMP`
Systems, Cloud & Tools	Linux, `AWS`, Git, Hugging Face, Docker, LaTeX

Awards and Fellowships

2026 **Texas A&M Institute of Data Science (TAMIDS) Student Travel Grant**, Texas A&M University

2026 **IMS Hannan Graduate Student Travel Award**, Institute of Mathematical Statistics

2026 **Section on Bayesian Statistical Science (SBSS) Student Paper Award**, American Statistical Association

2026 **Symposium on Data Science & Statistics (SDSS) Student and Early Career Scholarship**, American Statistical Association

2026 **Best of Statistical Science (BOSS) TidBit Award**, Texas A&M University

2026 **Graduate Research and Presentation (RAP) Travel Award**, Texas A&M University

2024 **NSF Travel Grant**, for presenting at IISA 2024

2024 **Targeted Proposal Teams (TPT) Grant**, Research Assistantship | Texas A&M University

2024 **Selected for CMS^3-FAST Summer School**, Texas A&M University

2022 **R. C. Bose Memorial Book Prize**, Calcutta Statistical Association

2022 **IASc-INSA-NASI Science Academies Summer Research Fellowship**, Indian Academy of Sciences

2022 **OPHI Summer School Grant**, University of Oxford

2022 **IAOS Conference and Travel Grant**, World Bank

Talks and Presentations

2026 **International Indian Statistical Association Conference** Student Paper Award Presentation — Varanasi, India
Probabilistic Symbolic Regression for Equation Discovery via Operator-induced and Regularized Symbolic Forests

2026 **Symbolic Artificial Intelligence and Machine Learning in Baixada Santista Workshop** Invited Talk — São Paulo, Brazil
Probabilistic Symbolic Regression for Equation Discovery via Operator-induced and Regularized Symbolic Forests

2026 **Joint Statistical Meetings** Invited SBSS Award Presentation — Boston, MA
Probabilistic Symbolic Regression for Equation Discovery via Operator-induced and Regularized Symbolic Forests

2026 **Symposium on Data Science & Statistics** Refereed Invited Talk — Milwaukee, WI
Hierarchical Bayesian Operator-Induced Symbolic Regression Trees for Structural Learning of Scientific Expressions

2025 **STAT CAFÉ** Seminar Talk — College Station, TX
Bayesian Symbolic Trees for Structural Learning of Scientific Expressions

2025 **SIAM Texas-Louisiana Sectional Meeting** Invited Talk — Austin, TX
Physics-Informed Neural Operators for Seismic Wave Modeling

2025 **Los Alamos National Laboratory Student Symposium** Poster — Los Alamos, NM
Physics-Informed Neural Operators for Seismic Wave Modeling

2024 **International Indian Statistical Association Conference** Contributed Talk — Kochi, India
Tangent Approximation for Variational Inference in Different Exponential Families

Leadership and Professional Activities

- **Workflow Workshop Organizer** — **2024–2025**
  Statistics Graduate Student Association, Texas A&M University
  Led and coordinated graduate workshops on research workflow, computational practice, and academic productivity.
- **Organizing Committee Member** — **2019–2020**
  Department of Statistics, St. Xavier's College (Autonomous), Kolkata
  Served on the organizing committee for the annual departmental fest ϵpsilon δelta, supporting event planning and coordination.

References

Bani K. Mallick
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